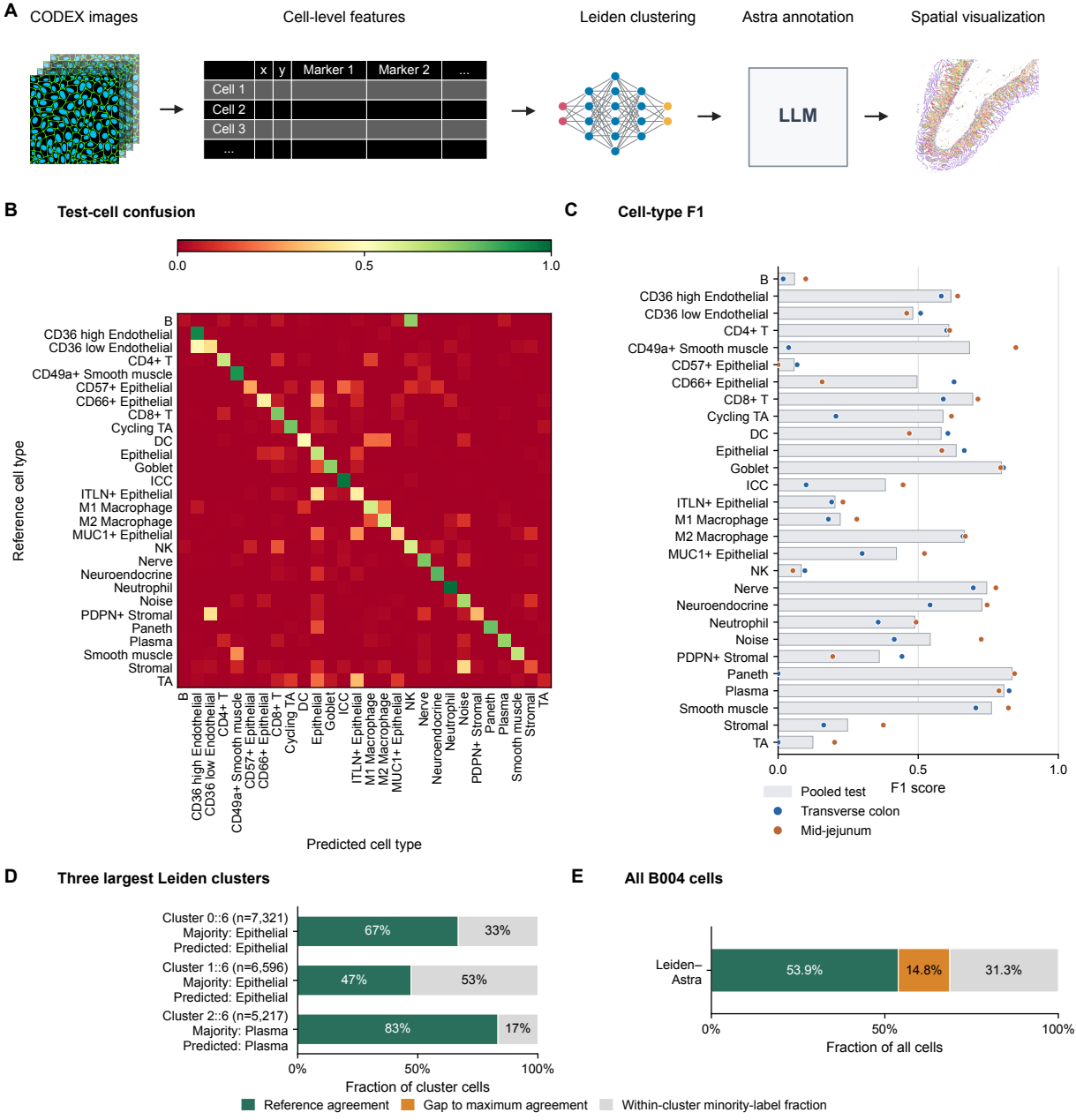

A
CODEX images
Cell-level features
Leiden clustering
Astra annotation
Spatial visualization
x
y
Marker 1
Marker 2
...
Cell 1
Cell 2
Cell 3
...
LLM
B
Test-cell confusion
0.0
0.5
1.0
Reference cell type
Predicted cell type
B
CD36 high Endothelial
CD36 low Endothelial
CD4+ T
CD49a+ Smooth muscle
CD57+ Epithelial
CD66+ Epithelial
CD8+ T
Cycling TA
DC
Epithelial
Goblet
ICC
ITLN+ Epithelial
M1 Macrophage
M2 Macrophage
MUC1+ Epithelial
NK
Nerve
Neuroendocrine
Neutrophil
Noise
PDPN+ Stromal
Paneth
Plasma
Smooth muscle
Stromal
TA
C
Cell-type F1
F1 score
Pooled test
Transverse colon
Mid-jejunum
D
Three largest Leiden clusters
Cluster 0::6 (n=7,321)
Majority: Epithelial
Predicted: Epithelial
67%
33%
Cluster 1::6 (n=6,596)
Majority: Epithelial
Predicted: Epithelial
47%
53%
Cluster 2::6 (n=5,217)
Majority: Plasma
Predicted: Plasma
83%
17%
Fraction of cluster cells
Reference agreement
Gap to maximum agreement
Within-cluster minority-label fraction
E
All B004 cells
Leiden–Astra
53.9%
14.8%
31.3%
Fraction of all cells